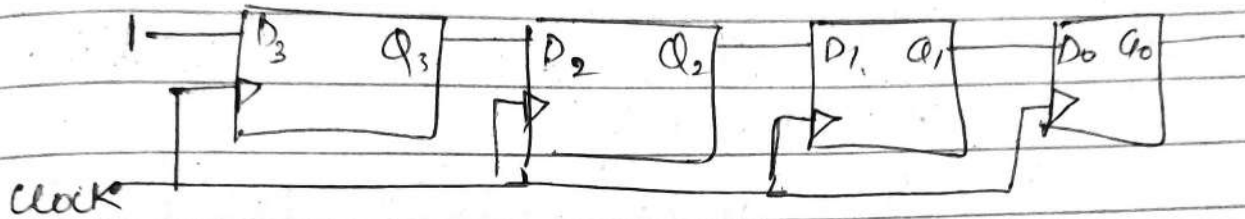


Shift Registers Part A Preena Agarwal

4 bit SISO

Given input sequence 1101 MSB First

after 4 cycle, output seq??



clock bit Register

0

0000
Q3 Q2 Q1 Q0

previous value is stored & no transfer
happens

↑

1 (MSB)

1000

↑

1

1100

↑

0

0110

↑

1

1011

output

2Q

initial stored 1010

Input Bits 1100

after 4 shift??

clock input Bit Register

0

1010

↑

1

1101

↑

1

1110

↑

0

0111

↑

0

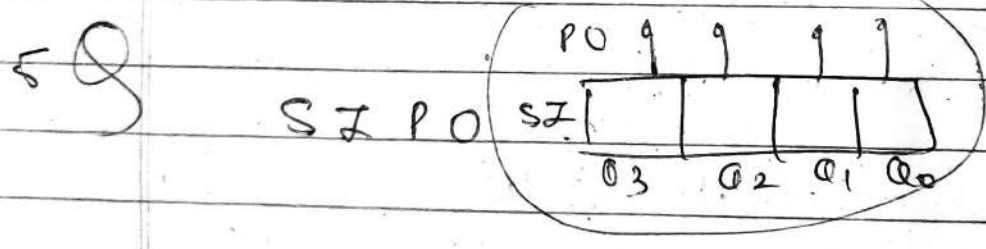
0011

3 Q $f_{clk} = 1\text{MHz}$ How long does it take
4 bit SZBO register to completely
output 1 data word

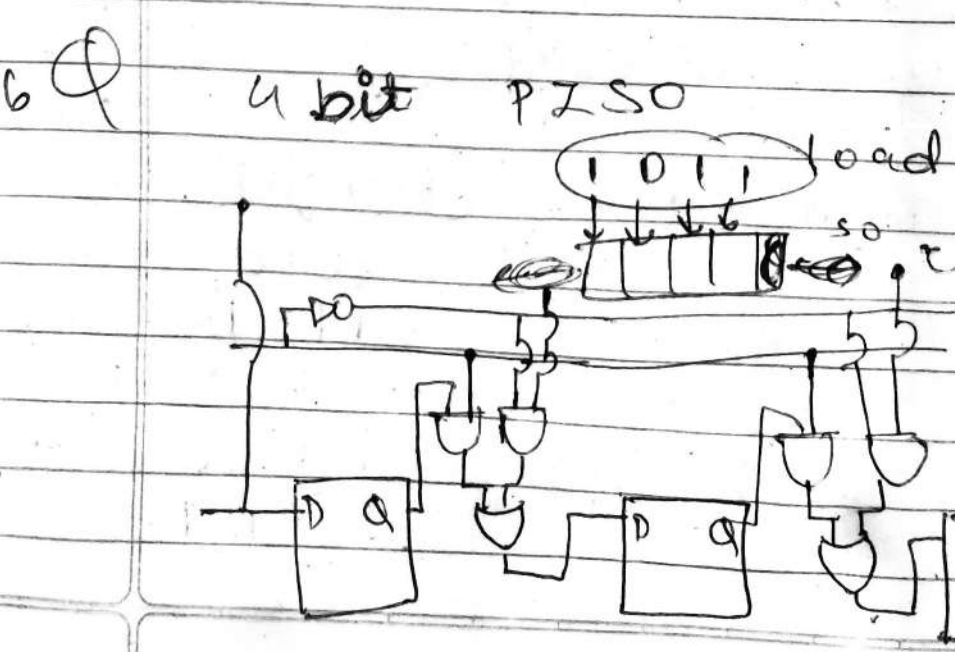
Need 4 cycles
 $4 \times \frac{1}{1\text{MHz}} = 4\mu\text{s}$

4 Q 4 bit SZPO
 input sequence 1001 (LSB first)
 output 2? after 4 clock.

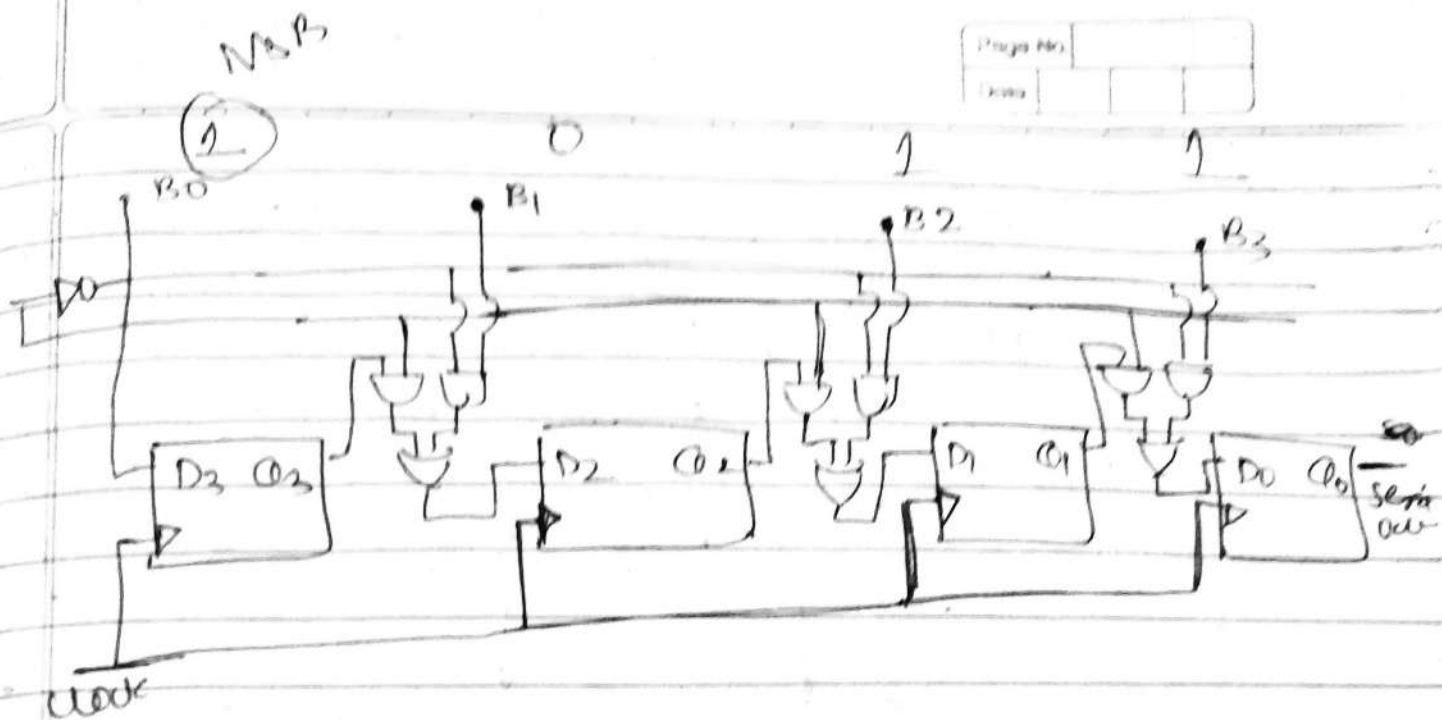
clock	input bit	output
↑	1	1000
↑	0	0100
↑	0	0010
↑	1	1001



8 Bit word
 serially load
 ignore the line
 8 clocks required



output sequence
 after 4 clock
 cycles



Serially out MSB first

1

0

1

1

1011 ✓

7 Q 4bit PISO register, load 1100
Shift dirⁿ is Right
output seq??

1100
serial dirⁿ Right ∴ LSB serial out

↑ 0

↑ 0

↑ 1

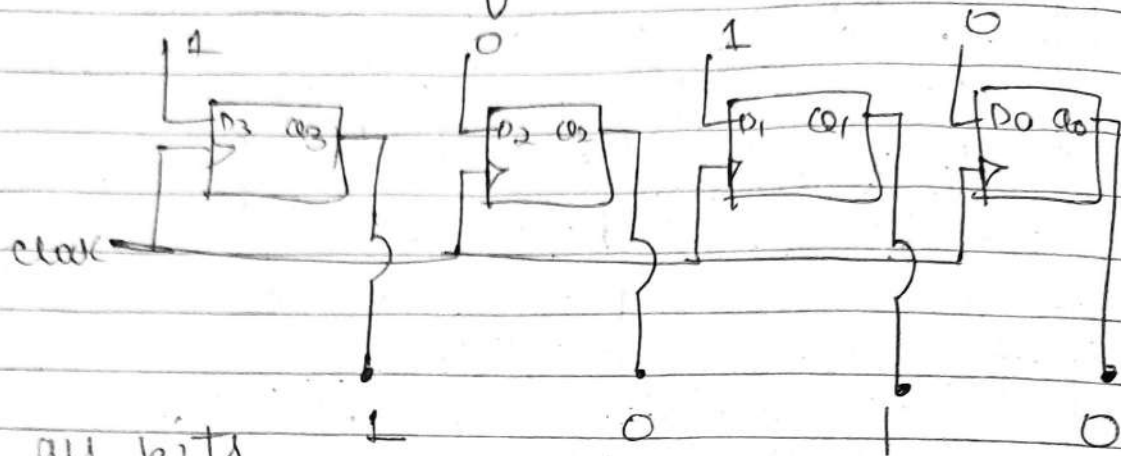
↑ 1

4 clock
cycles are
required

1100
0110
0011
0011

answer

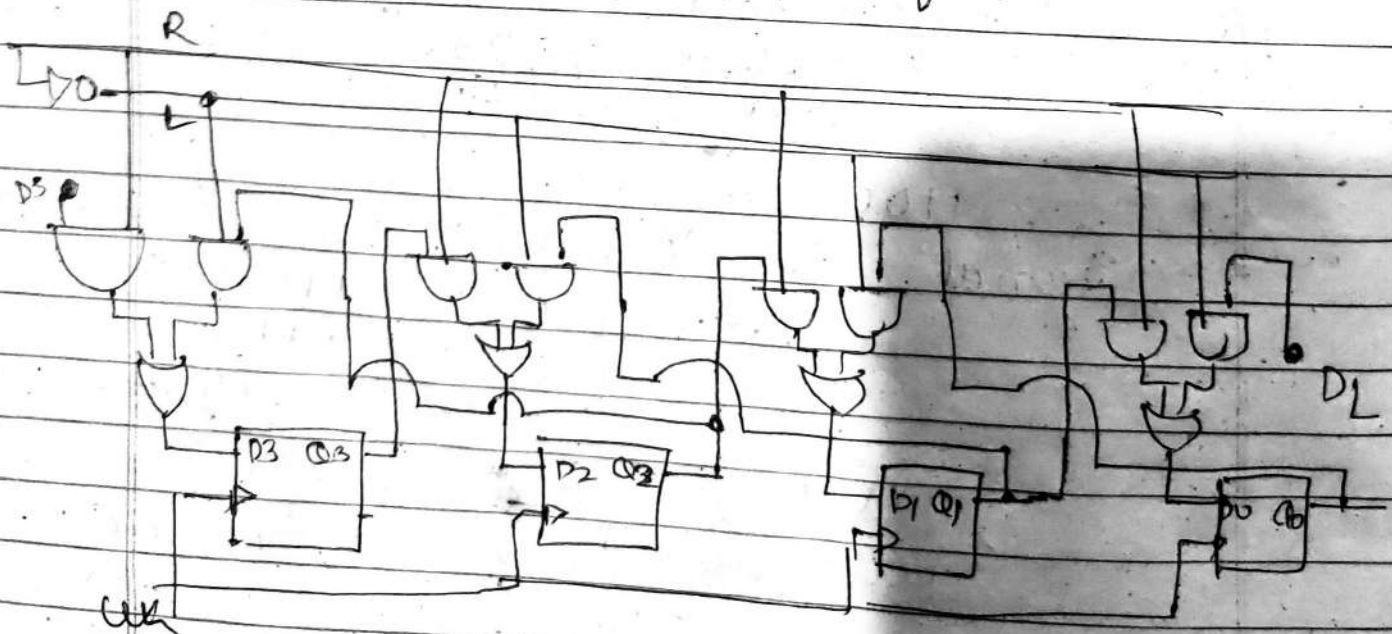
8 Q 4 bit PISO loaded with 1010
output after 1 clock pulse?



all bits
output simultaneously after 1 clock cycles.

9 Q How a bidirectional shift Register is used to implement a reversible counter?

→ A Memory circuit that moves stored binary data either left or right.



change
shift direction
using control
signal Left/Right

By controlling shift direction these allow counting sequence to proceed in increasing or decreasing order of count

Shift Right
↓
count down

Shift Left
↓
count up

108 Universal Shift Register

S_1 (MSB) S_0 (LSB)

0 0 (Hold)

0 1 Shift R C

1 0 Shift L

1 1 parallel load (P2PO & P2SO)

parallel load input 1101

if mode = shift Right & serial input = 1

Register content after 2 clock??

loaded (1101)

